What Makes Web Text Useful for Pretraining? Interpretable Document Quality from Filters, Latent Factors, and Model Swarms

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Abstract

We ask which interpretable properties of a web document predict its usefulness as language-model pretraining data. On a 2.26M-document subsample of `WebOrganizer/Corpus-200B` annotated with ten quality filters, global and within-topic Spearman correlations differ by less than 0.07 everywhere, so filter disagreement persists inside topics and is not explained away by topic mixture. A two-factor Gaussian measurement model fit by EM recovers two interpretable axes: a benchmark axis (ARC-Easy, PIQA, SciQ) and a generative-quality axis (DCLM, FineWeb-Edu, LAMBADA). Because that target is still a function of existing filter scores, we estimate an independent document-level contribution by training 593 small language models on randomized document subsets and applying difference-in-means to held-out validation loss. An Explainable Boosting Machine on 42 human-readable features (no perplexity-correlation scores) reaches test $R^2 = 0.656$ against a split-half noise ceiling of 0.57–0.70; adding the four perplexity-correlation scores raises this to 0.668, while a Lasso baseline reaches 0.408. In downstream training of 60M-parameter language models on equal 150M-token budgets, the full EBM applied within topics beats random selection on broad CommonCrawl validation loss; DCLM-fastText wins on ARC-Easy and LAMBADA. The best selection rule depends on the target metric and the diversity constraint.

1 INTRODUCTION

Pretraining data selection is a statistical decision problem at the scale of modern language modeling. Given a budget of tokens and GPU hours, the practitioner must choose which documents receive the next unit of compute. Existing pipelines answer this with classifier scores, heuristic thresholds, perplexity correlations, or domain mixtures (Li et al., 2024; Penedo et al., 2024; Thrush et al., 2025; Wettig et al., 2024, 2025). These tools work well in practice, but they do not directly answer the scientific question behind filtering: which interpretable document properties lower validation loss during training?

The starting point is a descriptive concern. WebOrganizer showed that organizing CommonCrawl into topic, format, and domain structure explains a large part of the gain of some filters (Wettig et al., 2025): about 84% of FineWeb-Edu's downstream gain is recovered from topic reweighting alone, versus only $\sim 35\%$ for DCLM-fastText. If a filter works mainly by changing the corpus mixture, calling its selected examples "higher quality" is misleading: it may simply choose more educational or technical topics. Conversely, if filters disagree after conditioning on topic, then the useful signal is genuinely document-level and cannot be recovered by domain reweighting alone.

The paper proceeds from descriptive structure to validated data selection. We do four things. (i) On the 2.26M-document subsample of `WebOrganizer/Corpus-200B` annotated with ten filter scores, global and within-topic Spearman correlations differ by less than 0.07 everywhere: filters disagree at the document level inside topics, not because they rank topics differently. (ii) A two-factor Gaussian measurement model

$$h_i \mid z_i \sim \mathcal{N}_2(\Gamma^\top z_i, I_2),$$
$$s_i \mid h_i \sim \mathcal{N}_K(\alpha + \Lambda h_i, \operatorname{diag}\psi),$$

recovers two interpretable filter axes by EM, and its posterior score defines a tunable quality–diversity policy. (iii) To replace the circular filter-against-filter target, we train 593 small language models on randomized document subsets and estimate a per-document contribution y_i by difference-in-means on held-out validation

loss. The split-half reliability of y is 0.40, giving a noise-limited R^2 ceiling of 0.57–0.70. (iv) An Explainable Boosting Machine fits y from 42 human-readable features to test $R^2 = 0.656$, close to that ceiling; trained 60M Llama models on equal 150M-token budgets show that the full EBM applied within topics is the only learned filter that beats random on broad CommonCrawl validation loss, while DCLM-fastText is best on benchmark-style tasks.

Code, scripts, and posterior artifacts are at `https://github.com/kkaitlyn111/WebOrganizer` (commit `d813bc7`).

2 DATA

2.1 Corpus and Annotations

We use `WebOrganizer/Corpus-200B` (Wettig et al., 2025), a 200B-token cleaned and deduplicated subset of CommonCrawl derived from the DataComps-LM `1b-1x` pool (Li et al., 2024), split into 9,888 shards. Each shard ships with aligned `.npy` annotations: DCLM-fastText scores, FineWeb-Edu scores, 24-category topic and format labels with logits, k-means cluster IDs, and GPT-NeoX token counts. We sample 100 of 9,888 shards (seed 42), yielding $D = 2{,}263{,}456$ documents.

Quality filters. We compile ten scores from four methodologically distinct families. (i) DCLM-fastText (Li et al., 2024), a bag-of-bigrams classifier. (ii) FineWeb-Edu (Penedo et al., 2024), a neural classifier distilled from Llama-3-70B educational-value ratings, available as rounded ordinal scores $\{0, \ldots, 5\}$. (iii) Four perplexity-correlation fastText filters (Thrush et al., 2025) targeting ARC-Easy, PIQA, SciQ and LAMBADA, each trained from 90-model correlations between language-model perplexity and benchmark performance. (iv) Four QuRater-1.3B axes (Wettig et al., 2024): writing style, required expertise, facts & trivia, and educational value. We also extract 16 Gopher/MassiveText-style heuristics (Rae et al., 2021; Raffel et al., 2020): character, word and line counts; mean word and line length; fractions of alphabetic, digit, punctuation, and uppercase characters; fractions of lines with terminal punctuation and bullet starts; type–token ratio; stopword fraction; and repeated n-gram coverage for $n \in \{2, 3, 4\}$.

For contribution modeling we use a 2.06M-document training pool after requiring the full feature panel. The full panel has 46 columns: four QuRater axes, a textbook-quality score, an AI-generation probability, entity density and diversity, entities per sentence, discourse and explanation markers, question density, pronoun and first-person fractions, numeric and proper-noun density, Flesch and Flesch–Kincaid readability, the 16 Gopher heuristics, and the four perplexity-correlation scores. We report two EBM variants: a *no-PC* EBM trained on the 42 human-readable features (excluding the four perplexity-correlation scores), and a *full* EBM on all 46. DCLM and FineWeb-Edu are excluded from both predictors and retained as downstream baselines.

2.2 Exploratory Findings

Score distributions are heterogeneous. DCLM is bimodal on $[0, 1]$ with kurtosis $\approx$ 80; perplexity-correlation scores pile up near the positive label; QuRater logits are close to Gaussian; FineWeb-Edu is discrete. The LAMBADA filter is right-skewed with mode near 0.1, essentially the mirror image of ARC-Easy, PIQA, and SciQ (modes $\sim$ 0.8–0.9). Bounded scores call for logit transforms (with Student-t errors for DCLM, given the heavy tails); the rest are modelled on their native scales after standardization. The URL-domain distribution is heavily Zipfian: 798,128 unique registered domains, $\sim$ 2,300 with $\geq$ 100 documents, $\sim$ 180 with $\geq$ 1,000.

Filters form interpretable but incomplete correlation blocks. The pairwise Spearman heatmap in Figure 1 (right) shows three blocks: ARC-Easy and PIQA at $\rho = 0.84$, with SciQ at 0.43–0.46 to both; LAMBADA almost orthogonal to the other perplexity-correlation filters ($\rho \leq 0.06$); DCLM at 0.40–0.41 rank-correlation with FineWeb-Edu and LAMBADA despite near-zero Pearson correlation, so the shared signal is concentrated in the tails. The difference between global and within-topic Spearman correlations is nearly zero everywhere ($|\Delta\rho| < 0.07$): conditioning on topic barely changes the pairwise structure.

Quality is also topic-relative. Figure 1 (left) plots raw DCLM against its within-topic percentile rank, coloured by topic. The result is 24 distinct curves rather than one monotone relationship. A document at the 90th DCLM percentile within "Adult" has a raw score well below the median of "Science & Technology". A global threshold therefore implicitly removes entire topics rather than selecting the best documents within each.

3 MODEL

3.1 Two-Factor Latent Measurement Model

A full multivariate hierarchical regression with topic, format, domain, and heuristic random effects plus a dense cross-filter residual covariance is useful descriptively, but it does not produce a per-document quality

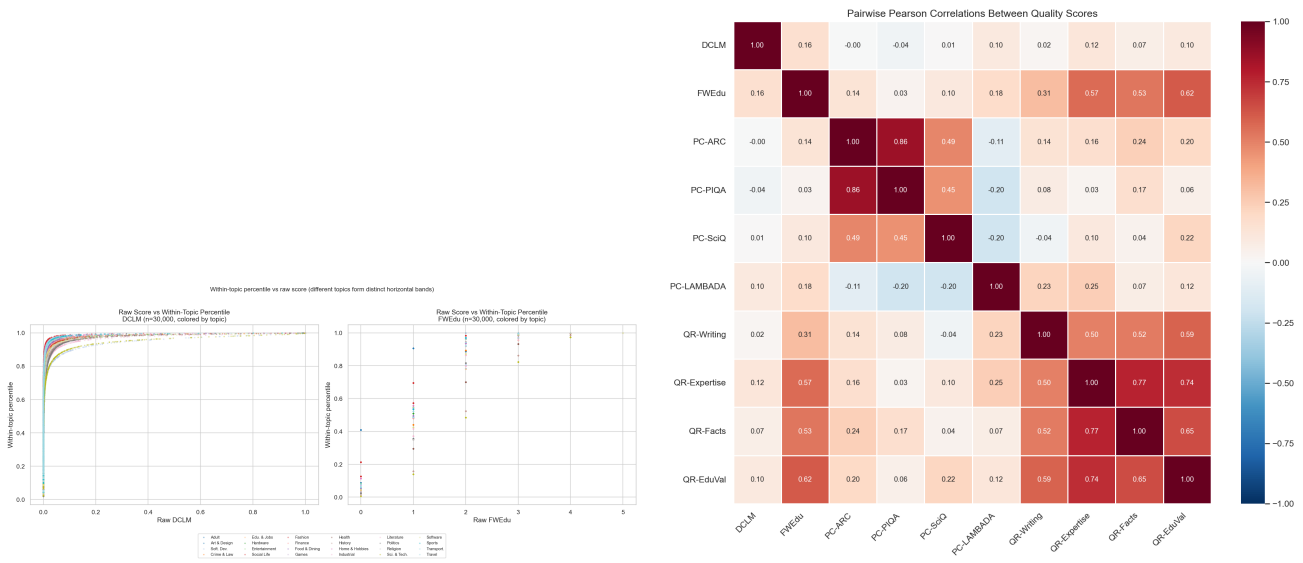


Figure 1: Exploratory evidence. Left: raw filter scores map to different within-topic percentiles by topic, so a global threshold implicitly changes the topic mixture. Right: pairwise Spearman correlations show three blocks: an ARC/PIQA/SciQ benchmark cluster, a DCLM/FineWeb-Edu/LAMBADA cluster, and an essentially orthogonal LAMBADA axis. Global and within-topic correlations differ by less than 0.07 everywhere, so filter disagreement persists inside topics.

ranking usable for selection and is too parameter-heavy for our pool. We therefore model the $K = 6$ complete transformed scores (DCLM, FineWeb-Edu, ARC-Easy, PIQA, SciQ, LAMBADA) on a $\sim$ 120k-document sample with a low-rank Gaussian factor structure.

Let $s_i \in \mathbb{R}^K$ collect the six scores for document i after clipping bounded values at 10^{-4}, applying logit, and standardizing. Let $z_i \in \mathbb{R}^q$ contain an intercept, 17 text heuristics, 24 topic indicators, 24 format indicators, and indicators for the top 500 URL domains plus an "other" bucket. We posit two latent document axes $h_i \in \mathbb{R}^2$ with a regression prior:

$$h_i \mid z_i, \Gamma \sim \mathcal{N}_2(\Gamma^\top z_i, I_2), \tag{1}$$

$$s_i \mid h_i, \alpha, \Lambda, \psi \sim \mathcal{N}_K(\alpha + \Lambda h_i, \operatorname{diag} \psi). \tag{2}$$

The diagonal ψ keeps filter-specific noise explicit, while Λh_i captures shared cross-filter signal. We use flat priors for α, Λ, optimize $\psi > 0$, and regularize Γ with a Gaussian ridge prior. The fit is therefore MAP for the covariate effects and maximum likelihood for the measurement parameters.

Two modelling choices follow from the EDA. We use two factors rather than one because a one-factor fit loads almost entirely on DCLM/FineWeb-Edu/LAMBADA and leaves the ARC/PIQA/SciQ block unexplained; the Spearman block structure in Figure 1 predicts this. The regression prior on h_i via Γ lets the latent factors borrow strength from observable document structure, so that we can score documents from features alone when filter scores are unavailable.

3.2 An Independent Document-Level Target

The model in (1)–(2) gives a probabilistic decomposition of filter scores into two interpretable axes, but its evaluation target is still a function of existing filter scores. We replace this target with an empirical contribution. For each randomized training run r let ℓ_r be held-out validation loss, and let $\mathcal{R}_i^+$ and $\mathcal{R}_i^-$ be the runs that include or exclude document i. We define

$$y_i = \frac{1}{|\mathcal{R}_i^-|} \sum_{r \in \mathcal{R}_i^-} \ell_r - \frac{1}{|\mathcal{R}_i^+|} \sum_{r \in \mathcal{R}_i^+} \ell_r, \tag{3}$$

so positive y_i means that including i tended to *lower* validation loss. This is the difference-in-means estimator from the datamodels literature (Ilyas et al., 2022; Engstrom et al., 2024; Imbens and Rubin, 2015), applied to randomized subset assignment rather than leave-one-out training.

The estimand is pragmatic rather than a permanent property of a document. It is the marginal contribution under the distribution of training subsets, model size, token budget, optimizer, and validation set used in the experiment. This conditionality is unavoidable in pretraining data selection: a document that lowers broad CommonCrawl perplexity may not help ARC-Easy, and rankings at 15M parameters may not transfer to frontier scale.

3.3 Predictive Model on y

Given y_i, we fit

$$y_i = f(x_i) + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma^2), \tag{4}$$

where x_i is the feature vector. We compare a cross-validated Lasso to an Explainable Boosting Machine (EBM) (Lou et al., 2013; Nori et al., 2019), a generalized additive model

$$\hat{f}(x) = \beta_0 + \sum_k f_k(x_k) + \sum_{j<k} f_{jk}(x_j, x_k),$$

fit by cyclic gradient boosting of shallow trees on each feature. The one-dimensional shape functions f_k and selected pairwise interactions are directly inspectable, so the model supports ranking and interpretation simultaneously.

4 INFERENCE

4.1 EM for the Two-Factor Latent Model

The complete-data log-likelihood of (1)–(2) is Gaussian, so EM is exact. The E-step posterior is

$$p(h_i \mid s_i, z_i) = \mathcal{N}(m_i, V), \tag{5}$$

$$V = (I + \Lambda^\top \operatorname{diag}(\psi)^{-1}\Lambda)^{-1}, \tag{6}$$

$$m_i = V\{\Gamma^\top z_i + \Lambda^\top \operatorname{diag}(\psi)^{-1}(s_i - \alpha)\}. \tag{7}$$

V does not depend on i, so one Cholesky factorization suffices per iteration. The M-step has closed forms: α, Λ are updated by least squares using $\mathbb{E}[h_i] = m_i$ and $\mathbb{E}[h_i h_i^\top] = m_i m_i^\top + V$; ψ_k is updated as the residual variance of filter k given $s_i - \alpha - \Lambda m_i$ with the second-moment correction; and Γ is updated by ridge regression of $\mathbb{E}[h_i]$ on z_i, equivalent to a Gaussian prior on the covariate effects.

EM converged in 180 iterations. The approximate observed-data log-likelihood increased monotonically from $-746{,}210$ to $-678{,}610$, with a final increment of 0.067. We stop at increment < 0.1, so the optimizer terminated by tolerance rather than iteration cap. The factor structure is identifiable only up to a column permutation and a per-column sign flip, which is the expected class for an orthogonal-prior factor model. Held-out covariate-only prediction has RMSE 0.898; conditioning on the observed filters drops reconstruction RMSE to 0.625.

4.2 Subset Construction for the Swarm Target

We designed 791 subsets across five families: random thresholds (~ 200 runs, e.g., top-15% by entity density), multi-feature intersections (~ 300), linear combinations with random Gaussian weights (~ 200), targeted single-filter baselines (~ 50), and topic-conditional selections (~ 50). After enforcing $\geq$ 150M unique tokens per strategy, 593 subsets remained. Each run trained a 15M-parameter language model on 150M unique tokens for $\sim$ 20 minutes on an A100 (Figure 2, left). On average each document appears in 50.6 runs.

The design balances two goals that conflict in practice. Causal identification wants diverse, non-collinear inclusion patterns so that each document is not always bundled with the same neighbors. Practical data selection wants interventions that resemble real filters: selecting by one feature, by intersections, by a weighted score, or within a topic. The design estimates the contribution of each document under this realistic family of selection policies, not under arbitrary perturbations.

Noise ceiling. The split-half Pearson reliability of y across half-samples of runs is $\rho = 0.40$. By the Spearman–Brown formula this implies a disattenuated true-score correlation of $\rho^* = 2\rho/(1+\rho) \approx 0.57$ and an R^2 ceiling of 0.57–0.70 for any feature-based predictor.

5 RESULTS

5.1 Two-Factor Posterior

Figure 3 (left) shows the MAP factor loadings. Factor 1 is a benchmark axis: ARC-Easy loads 0.87, PIQA 0.80, SciQ 0.47. Factor 2 is a generative-quality axis with DCLM, FineWeb-Edu, and LAMBADA loadings 0.27, 0.27, 0.23. The unique variances ψ_k are smallest for ARC-Easy (0.064) and PIQA (0.186); DCLM, FineWeb-Edu, SciQ, and LAMBADA retain more filter-specific noise. There is not one universal quality filter, but two consistent latent axes. On held-out documents, Factor 1 correlates 0.71 with the composite six-score target and Factor 2 correlates 0.45.

Posterior data-mix policy. The posterior expected reconstructed score $\mathbb{E}[\alpha + \Lambda h_i \mid s_i, z_i]$ is a one-dimensional ranking we treat as a data-mix policy. Table 1 compares top-20% selection policies on held-out documents using the average of the six standardized scores as composite quality. Random selection has mean quality 0.001. Global DCLM reaches 0.405 with topic entropy 0.957. A one-factor posterior policy improves to 0.448. The two-factor posterior policy reaches 0.680, and the two-factor topic-floor variant maintains 0.638 quality while covering all 24 topics with entropy 0.964. A 1000-resample bootstrap gives a two-factor topic-floor improvement over DCLM of 0.274 standardized units, 95% CI $(0.261, 0.291)$. The composite target remains a function of the modelled

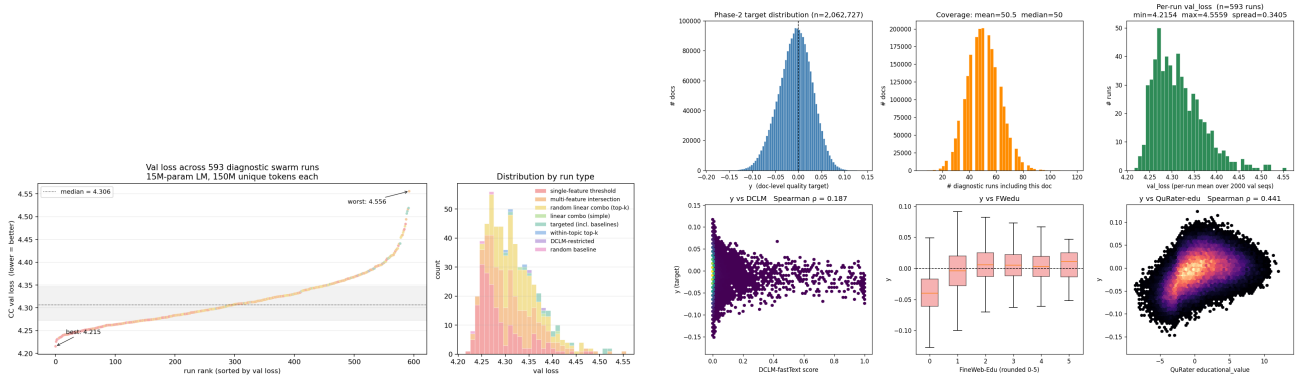


Figure 2: Target construction. Left: validation loss across the 593 trained diagnostic runs after token-budget filtering, sorted and coloured by subset family; dashed line marks the random-subset baseline. Right: distribution of the difference-in-means target y over $D = 2.06$M documents (top-left), per-document run coverage (mean 50.6, top-middle), per-run validation loss across the 593 swarm runs (top-right), and scatters of y against three example filters (bottom). Filters explain at most $\rho = 0.44$ of y individually.

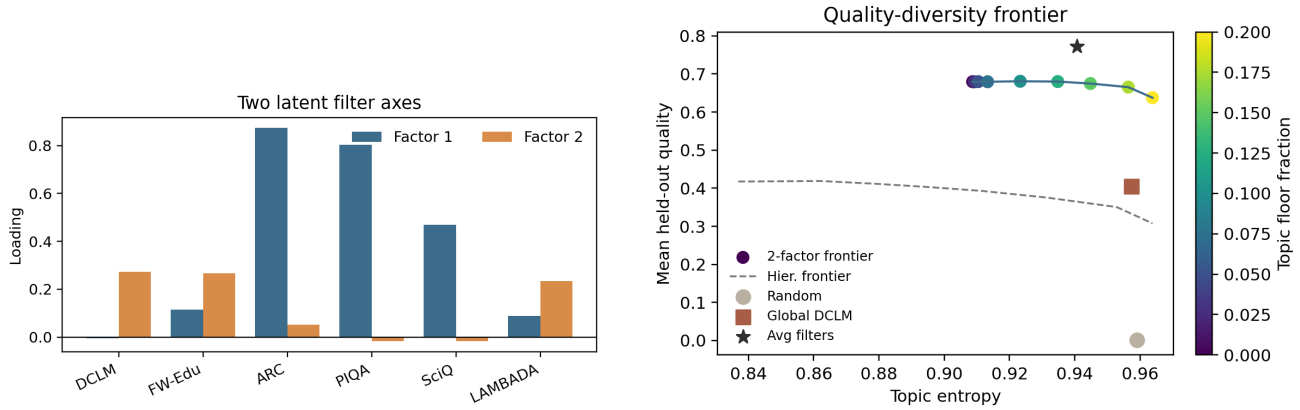


Figure 3: Posterior summaries. Left: two-factor MAP loadings Λ. Factor 1 (blue) loads on ARC, PIQA, SciQ; Factor 2 (orange) loads on DCLM, FineWeb-Edu, LAMBADA. Right: quality–diversity frontier of the two-factor posterior policy as the within-topic floor varies. The best frontier point is near a 10% floor (quality 0.681, entropy 0.923); the default for distributional safety is the 20% floor.

Policy	Quality	Entropy
Random	0.001	0.999
Global DCLM	0.405	0.957
One-factor posterior	0.448	0.952
Two-factor posterior	**0.680**	0.851
Two-factor (20% topic floor)	0.638	0.964

Table 1: Top-20% selection policies under the filter-composite target. The two-factor posterior dominates DCLM and the one-factor variant.

filters, so the downstream test in Section 5.3 is the more rigorous comparison.

5.2 Predicting Independent Contribution

With the difference-in-means target y from Section 4.2, we fit Lasso and EBM on an 80/20 stratified train/test split ($\sim$ 412,000 test documents). Table 2 reports train and held-out test R^2. Topic membership alone (ANOVA) explains 7.6% of the variance. Lasso on all 46 features reaches test $R^2 = 0.408$. The 42-feature human-readable EBM (no perplexity-correlation scores) reaches 0.656; adding the four perplexity-correlation scores raises this only to 0.668. Train and test R^2 are nearly identical for both EBMs, consistent with hitting the intrinsic noise ceiling of y rather than overfitting. The interpretable-only EBM therefore captures almost all of the available signal from human-readable text features.

Model	Train R^2	Test R^2	Features
Lasso (linear)	0.410	0.408	46 full
EBM, no-PC	0.658	0.656	42 readable
EBM, full	**0.671**	**0.668**	46 full

Table 2: Predicting the document-contribution target. PC = the four perplexity-correlation scores. The human-readable-only EBM reaches 99% of the full-feature EBM's test R^2.

Calibration and shapes. The binned test-set mean of y given $\hat{y}$ tracks the diagonal across the prediction range (Figure 4, right), with the largest uncertainty in the extremes where fewer documents live. Ranked by mean absolute contribution to $\hat{y}$, the top features are QuRater educational value (↑, importance 0.0059), textbook-quality score (∩ peaked), proper-noun density (↓ at high values), mean word length (↑), fraction alphabetic (↑), first-person sentence fraction (↓), QuRater required expertise (∩), Flesch reading ease (↓), Flesch–Kincaid grade level (∩), mean line length (↑), explanation marker density (↓), and QuRater facts & trivia (∩). Figure 5 shows the 1-D shape functions for the top six. QuRater educational value is the strongest signal and is largely monotone positive. Textbook quality peaks near 0.5 and then saturates: more textbook-like is not always better. First-person fraction and proper-noun density have steep negative regions at their high ends, flagging entity dumps and diary-style content as low-value.

5.3 Downstream Validation

We validate the EBM as a selection policy. We download 30 fresh CommonCrawl shards (612K documents, 540M tokens) disjoint from every earlier phase. Each policy selects 150M unique tokens; we train a 60M-parameter Llama model for 2 epochs (300M total tokens) and evaluate on held-out CommonCrawl perplexity (CC val), HellaSwag (Zellers et al., 2019), ARC-Easy (Clark et al., 2018), and LAMBADA loss (Paperno et al., 2016).

The full EBM applied within topics is best on CC val (3.873 nats), beating random (3.884) and ahead of every other filter. It also wins HellaSwag (0.320). DCLM-fastText wins ARC-Easy (0.379) and LAMBADA (4.350), as one would expect because DCLM was trained against benchmark-like positives. FineWeb-Edu falls in the middle. Two seeds of the global full EBM (s42 row labelled *EBM full* and s43) differ by 0.004 nats on CC val, so the within-topic gain is larger than the seed-to-seed noise. Every global top-k filter (DCLM, FineWeb-Edu, full EBM, no-PC EBM, Lasso) loses to random on broad CC val by 0.02–0.13 nats. This is the "data-quality illusion" (Nait Saada et al., 2025): training on a narrow high-quality subset hurts performance on the broad held-out distribution. The within-topic variant uses the same full EBM score but applies the top-k rule inside each of the 24 topics; because filter disagreement is document-level inside topics, ranking within a topic recovers structure that global thresholds erase, and the topic mixture is preserved.

Policy	CC↓	HS↑	ARC↑	LAM↓
Random (s42)	3.884	.313	.349	4.528
Random (s43)	3.887	.315	.332	4.564
Random (s44)	3.896	.311	.340	4.501
EBM full within-topic	**3.873**	**.320**	.316	4.724
EBM full	3.914	.308	.337	5.168
EBM full (s43)	3.918	.311	.326	5.212
EBM no-PC	3.921	.307	.326	5.169
Lasso	3.960	.303	.370	5.307
DCLM-fastText	4.006	.304	**.379**	**4.350**
FineWeb-Edu	3.967	.308	.365	4.746

Table 3: Downstream validation. Each policy selects 150M unique tokens; 60M Llama trained for 2 epochs. Lower is better for CC val and LAMBADA, higher for HellaSwag (HS) and ARC-Easy (ARC). Bold = best per column.

5.4 Implications for Synthetic Data

The EBM shape functions in Figure 5 can be read as feature targets for synthetic generation. Synthetic corpora such as TinyStories (Eldan and Li, 2023), textbook-style Phi data (Gunasekar et al., 2023), and Synthetic Continued Pretraining (Yang et al., 2025) already control vocabulary, exposition style, and entity relationships; the shape functions add measurable targets: high educational value, moderate (not maximal) textbook quality, named-entity coverage sufficient for grounding without becoming a list, clean alphabetic text, limited first-person voice. The transfer is untested. A direct check is to score generated corpora with the same feature pipeline, compare their distributions to high-$\hat{y}$ web documents, and train equal-token models on synthetic subsets selected by the full EBM within topics.

6 DISCUSSION AND LIMITATIONS

The analysis succeeds when the target becomes external to the filters. The EDA shows that filters disagree in a structured way, and the factor model gives that structure a probabilistic interpretation, but the latent-factor posterior can only denoise and rebalance existing filter scores. The model-swarm target changes the problem: y_i measures an empirical contribution to validation loss under randomized training subsets. Once this target exists, the EBM can ask which readable document properties predict contribution, and within-topic ranking

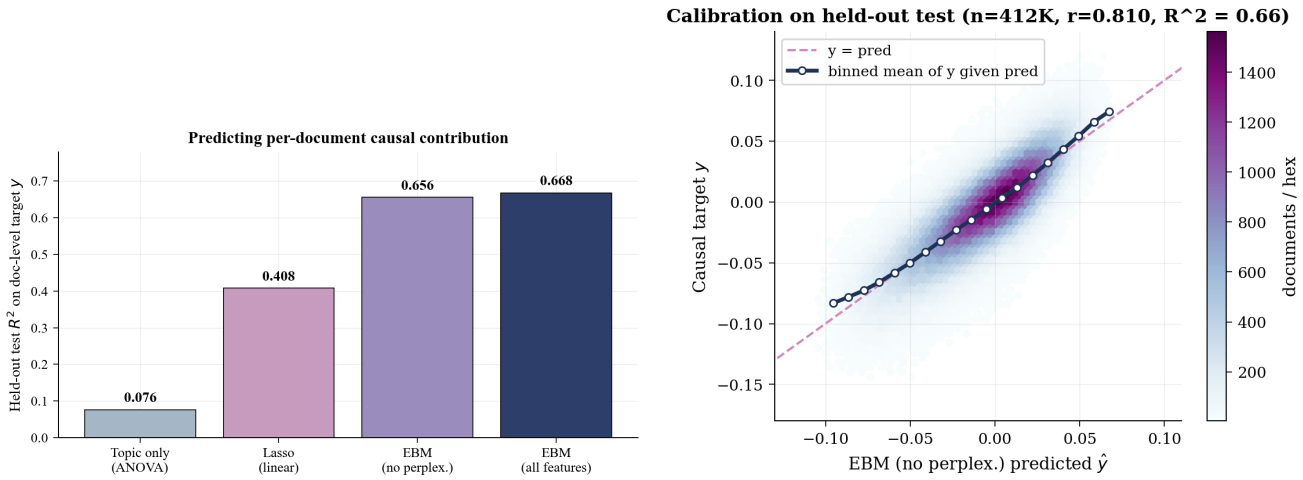


Figure 4: Predictor performance. Left: held-out test R^2 for four predictors of y. The no-PC EBM (42 human-readable features) sits just below the full-feature EBM and close to the 0.57–0.70 noise ceiling. Right: EBM calibration on the held-out test set ($n = 412$K, $r = 0.810$, $R^2 = 0.66$). Binned means of y given $\hat{y}$ track the diagonal.

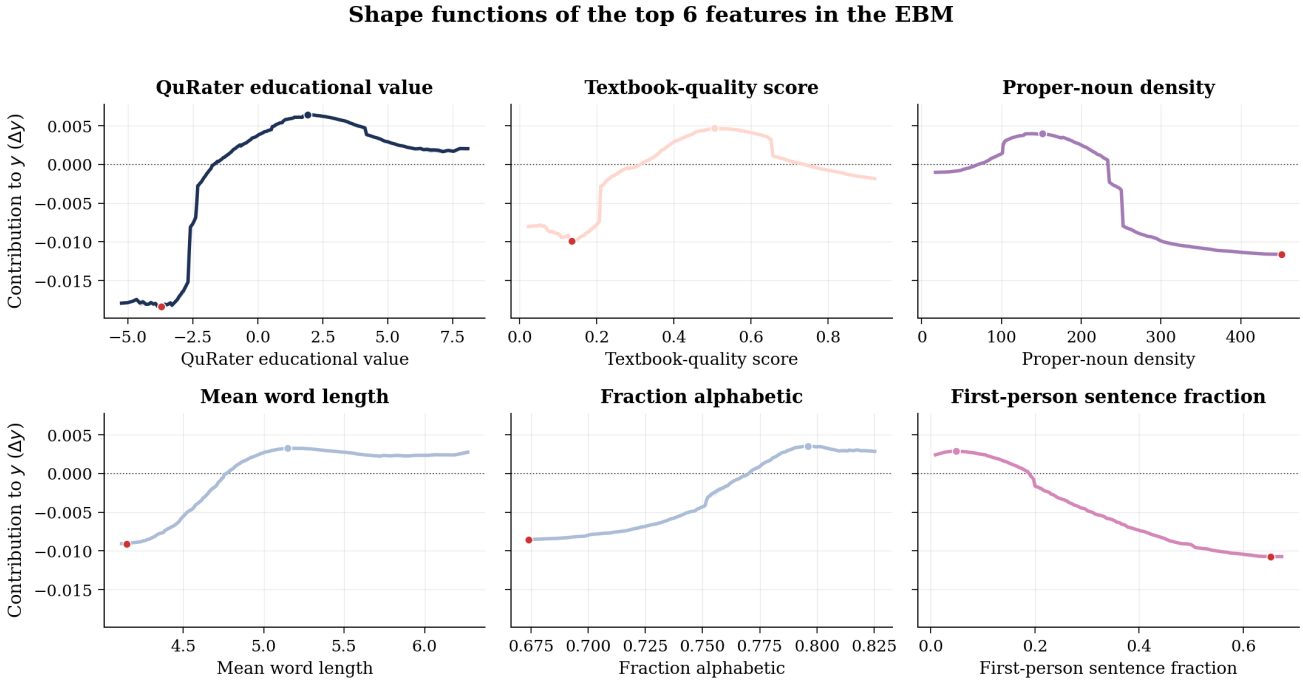


Figure 5: EBM 1-D shape functions for the top six features with 95% bag-bootstrap bands. Coloured dots mark the argmax; red dots mark the argmin.

translates that prediction into a selection policy that beats random on broad CC val.

A one-factor latent model is strictly worse than the two-factor variant: it collapses the ARC/PIQA/SciQ benchmark axis into the DCLM/FineWeb-Edu/LAMBADA axis and loses 0.23 in held-out composite quality on the top-20% policy. We also tested Grosse-style influence functions (Grosse et al., 2023) as an alternative document-level target: a 100M model trained on the full 1.8B-token pool gives one EK-FAC influence score per document, but the held-out R^2 against y was only ≈ 0.2 at this scale, so influence functions at our model size are not a substitute for randomized swarms.

The largest limitation is scale. y is built from 15M-parameter swarm runs, and validation uses 60M-parameter models on 300M tokens. Rankings may

reorder at the 1B+/25B-token scale at which DCLM was originally evaluated. The target is also noisy: split-half reliability 0.40 caps any predictor's possible R^2. The subset families that drive y are skewed toward random and single-feature selections, which account for ≈ 250 of the 593 runs; a design matrix with more multi-feature intersections would tighten identification. Broad CC val is one metric among many, and the conclusion that no filter wins on every metric is in part a consequence of evaluating on metrics whose distributions diverge. Finally, the EBM is fit on web documents; whether its shape functions transfer to synthetic corpora remains open.

7 REPRODUCIBILITY

The public repository is `https://github.com/kkaitlyn111/WebOrganizer`; the inspected commit is `d813bc7`. The codebase contains pipelines for annotation loading, feature extraction, factor-model fitting and EM, subset construction, model-swarm training, EBM fitting, fresh-validation data preparation, and downstream evaluation. The figures and validation numbers in Table 3 are reproduced from local experimental artifacts produced by that pipeline.

8 RESPONSIBLE AI USE

AI coding assistants, including Codex and Claude Code, were used for code generation, debugging, figure organization, and drafting assistance. For this manuscript, those assistants helped convert existing notes, repository outputs, and figures into AISTATS-format LaTeX. Scientific decisions, interpretations, and responsibility for correctness remain with the authors. Because the corpus is web text, data selection can amplify CommonCrawl biases; topic-aware validation reduces silent topic deletion but is not a fairness guarantee.

Acknowledgements

We thank the authors of WebOrganizer for releasing the corpus and annotations on which this paper is built.

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